

Total No. of Questions : 6]

SEAT No. :

P5778

[Total No. of Pages : 2

TE/Insem./Oct.-35
T.E. (Computer)
THEORY OF COMPUTATION
(2012 Pattern) (Semester - I)

Time :1 Hour]

[Max. Marks :30

Instructions to the candidates:

- 1) *Answer three questions.*
- 2) *Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data wherever required.*

Unit-I

- Q1)** a) Determine a regular expression over the alphabets {x, y} for the following, [4]
- i) All String containing exactly two x's
 - ii) All String that start with xy
- b) Construct a DFA over an alphabet {0, 1} for accepting the strings having number of l's as multiple of 3 [6]

OR

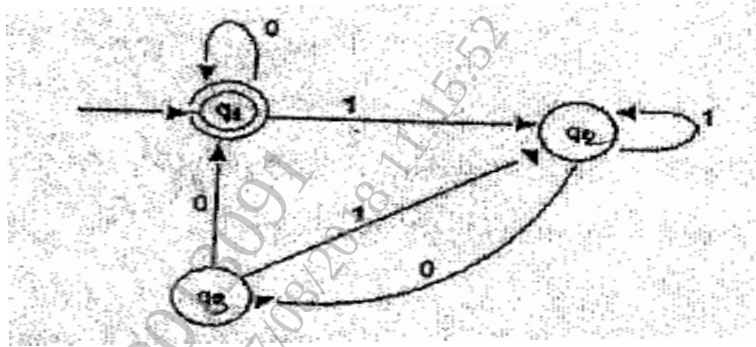
- Q2)** a) What is Kleen Closure? What is Positive closure? For a given language L under what Circumstances will L^+ and L^* equal? [6]
- b) Using pumping lemma for regular Sets provide that the language, $L = \{0^m 1^n 0^{m+n} | m \geq 1 \text{ and } n \geq 1\}$ is not regular. [4]

Unit-II

- Q3)** a) Differentiate between DFA and NFA with suitable example. [2]
- b) Find the Regular Expression for the set of strings recognized by the given FA. [4]

P.T.O.

Use Arden's Theorem.



- c) Construct the DFA for the language of all string that begin and end with same symbol Over the alphabet $\Sigma = \{a, b\}$. [4]

OR

- Q4)** a) Design a Moore machine for computing the 2's complement of a binary number. Convert it into its equivalent Mealy machine. [6]
 b) Explain the Closure and Decision properties of finite Automata. [2]
 c) Explain limitations of Finite Automata. [2]

Unit - III

- Q5)** a) Write a context free grammar for the following language $0(0+1)^* 01(0+1)^* 1$. [4]
 b) For the Left linear grammar below, obtain the equivalent Right linear grammar. [6]

$S \rightarrow C0|A0|B1$
 $A \rightarrow A1|C0|B1|0$
 $B \rightarrow B1|1$
 $C \rightarrow A0$

OR

- Q6)** a) Explain with suitable examples, any two applications of Context Free Grammars. [4]
 b) Convert the following CFG into Chomsky Normal form (CNF) [6]

$S \rightarrow ASA|aB$
 $A \rightarrow B|S$
 $B \rightarrow b|\epsilon$

